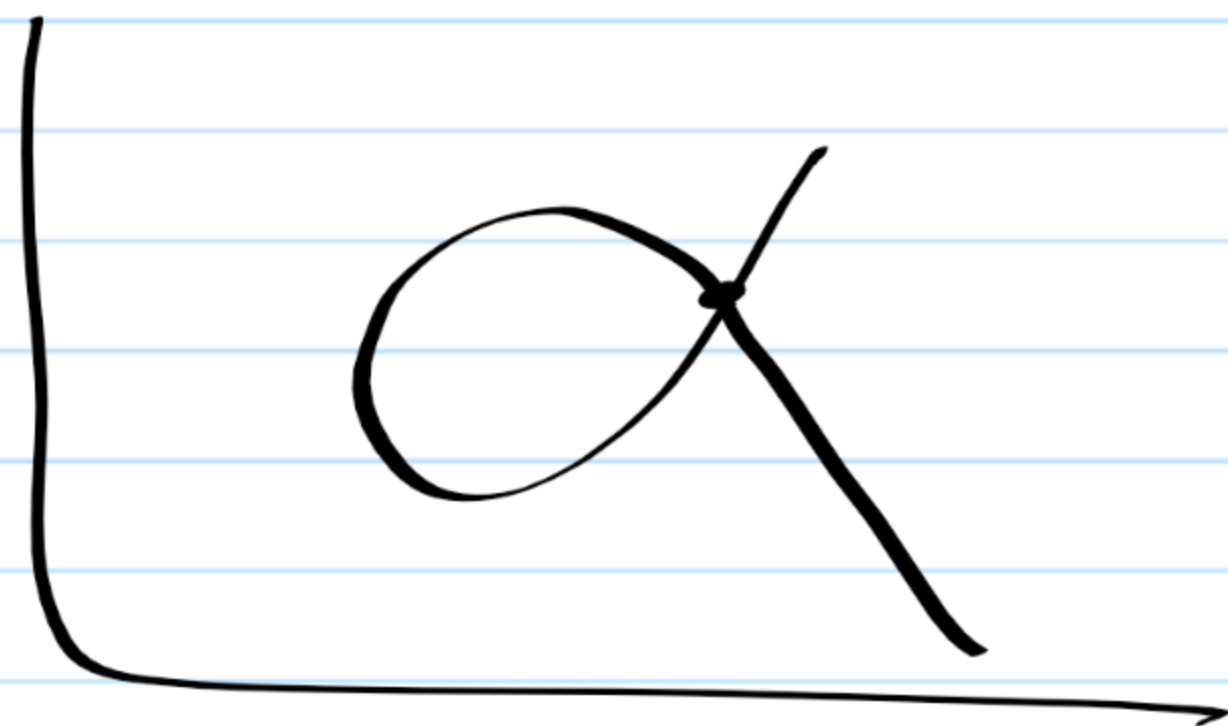
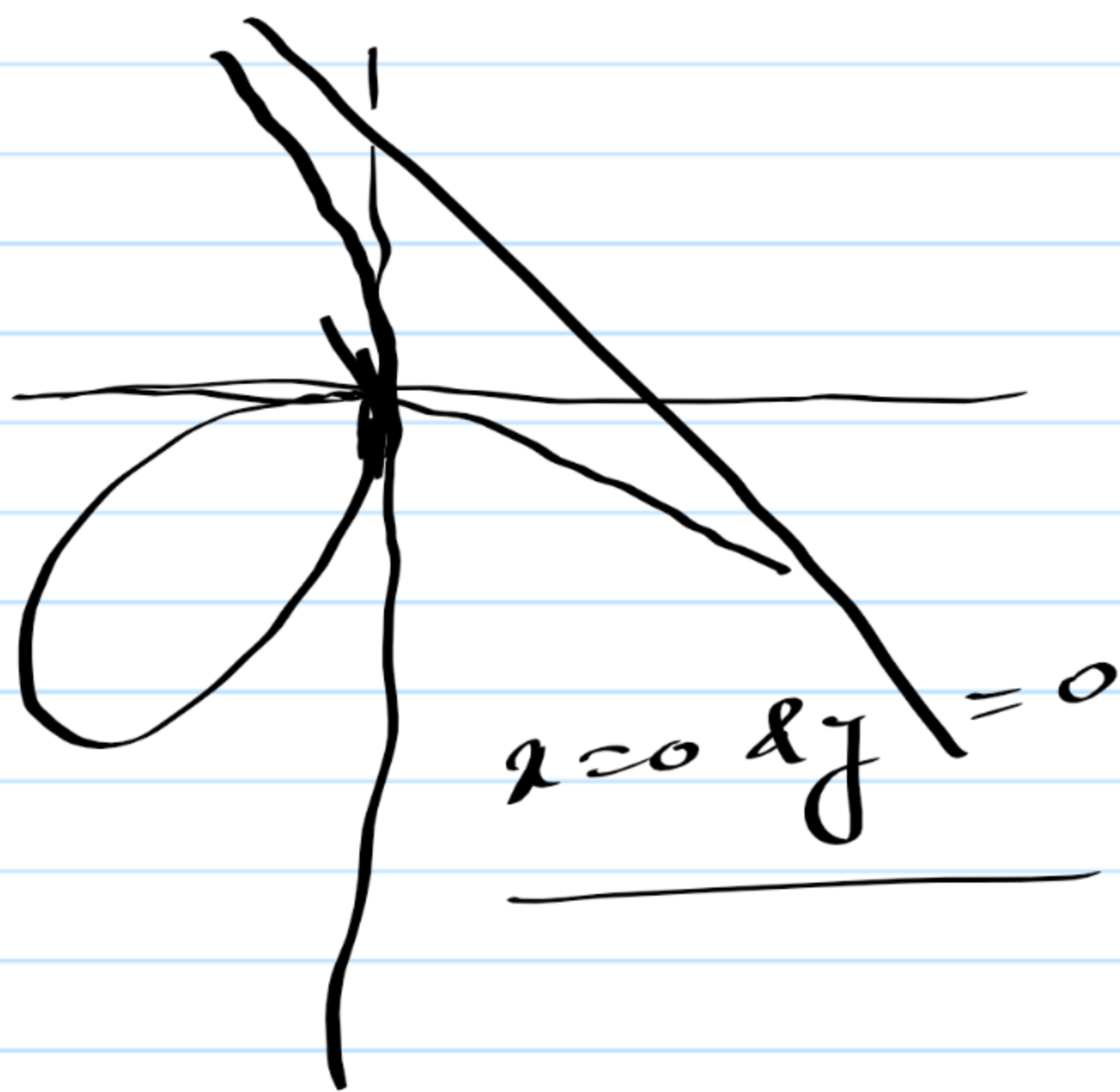


Singular & nonsingular pts.

A curve with no singular pts is called a nonsingular curve.

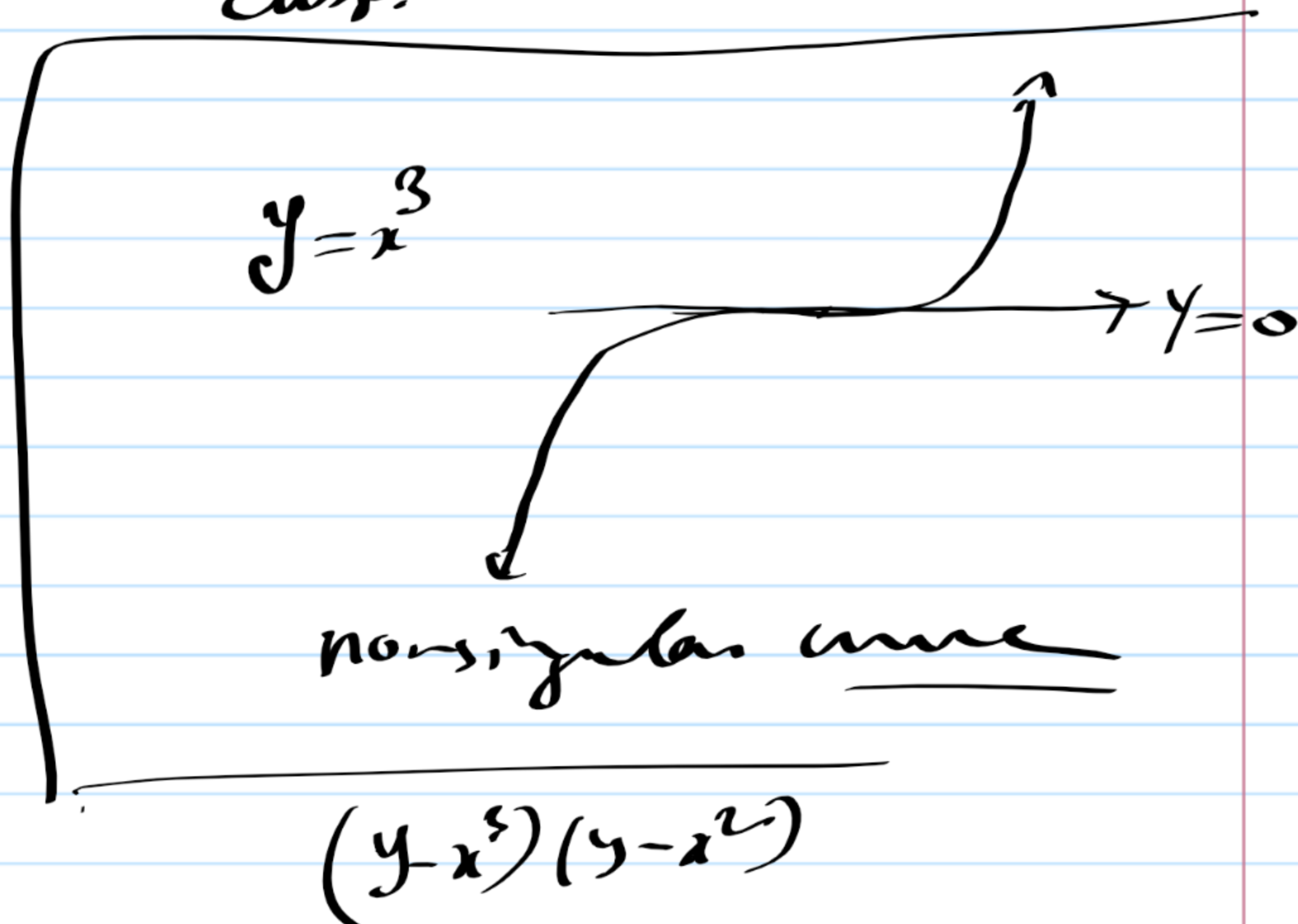
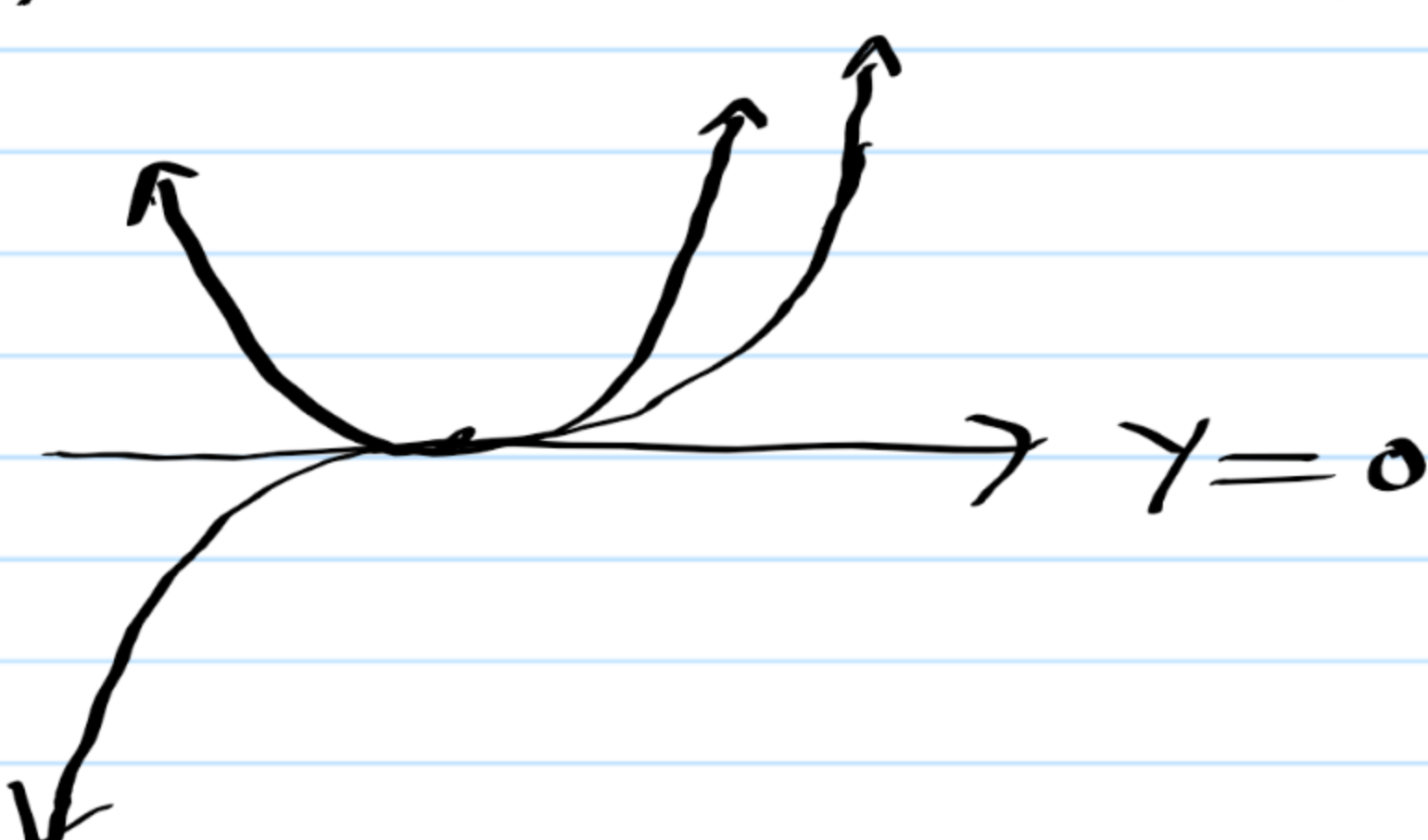
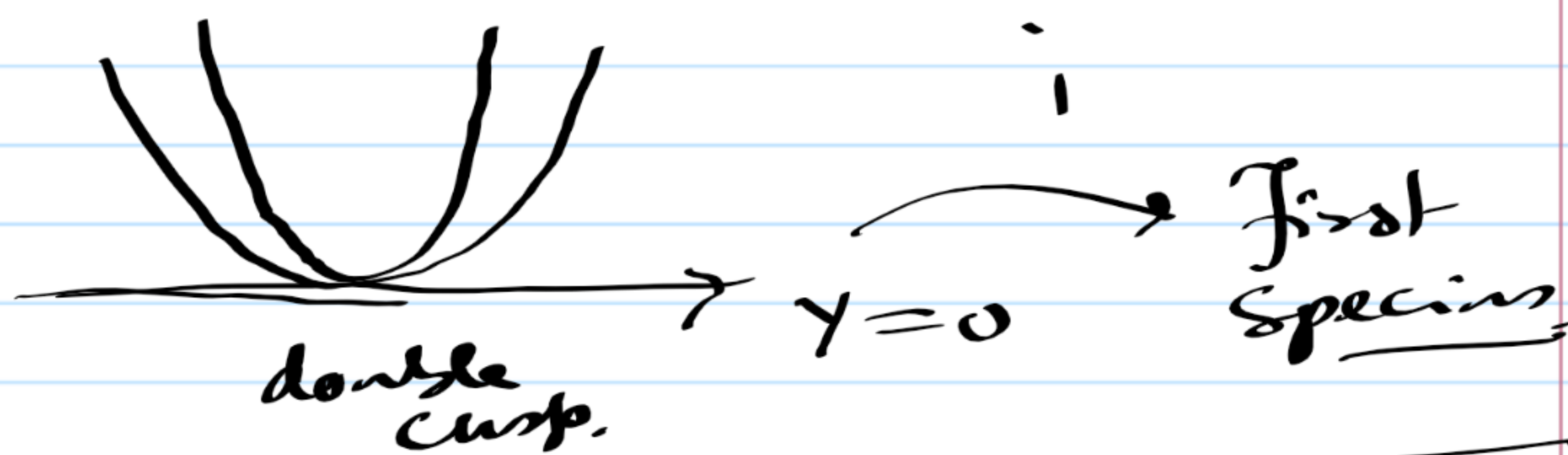
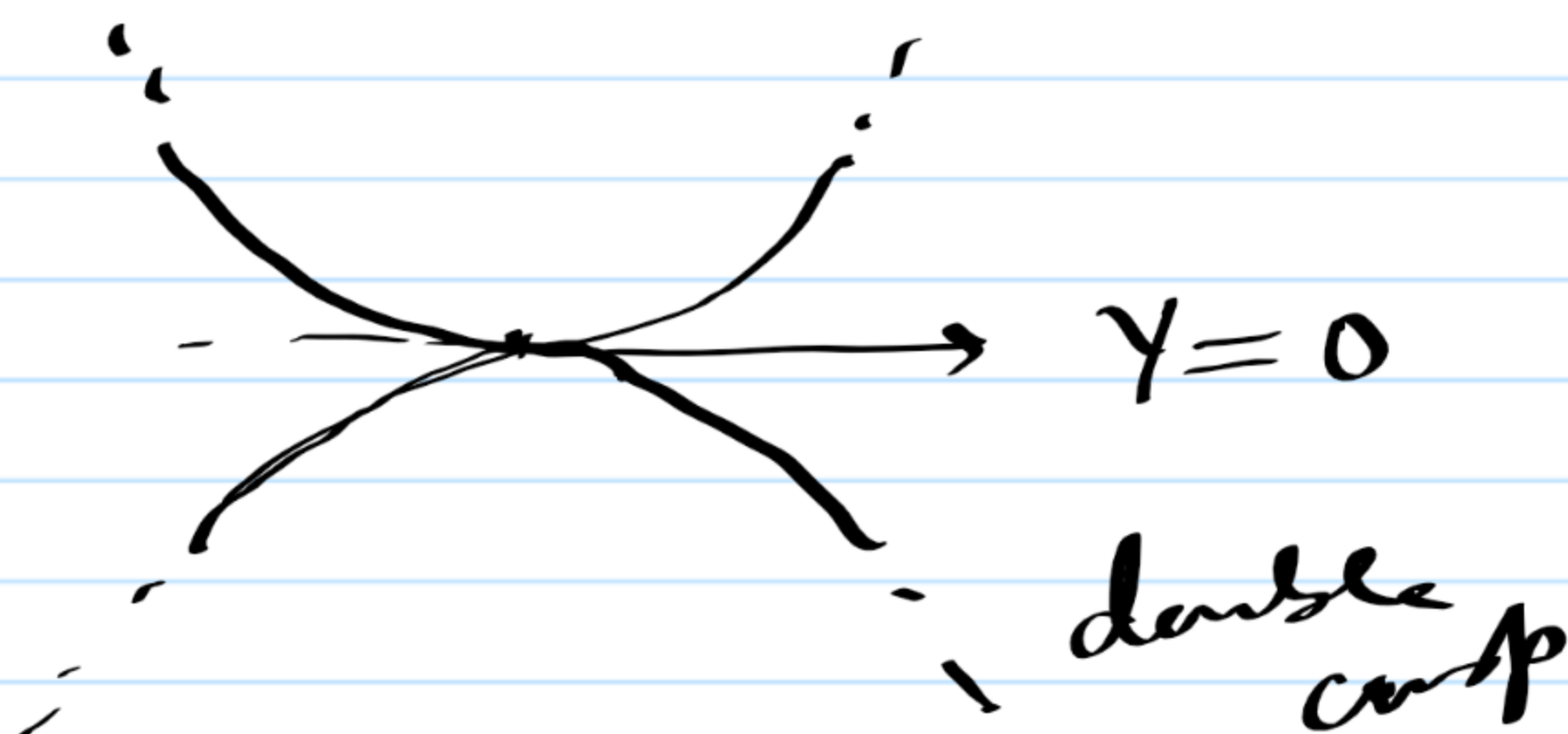
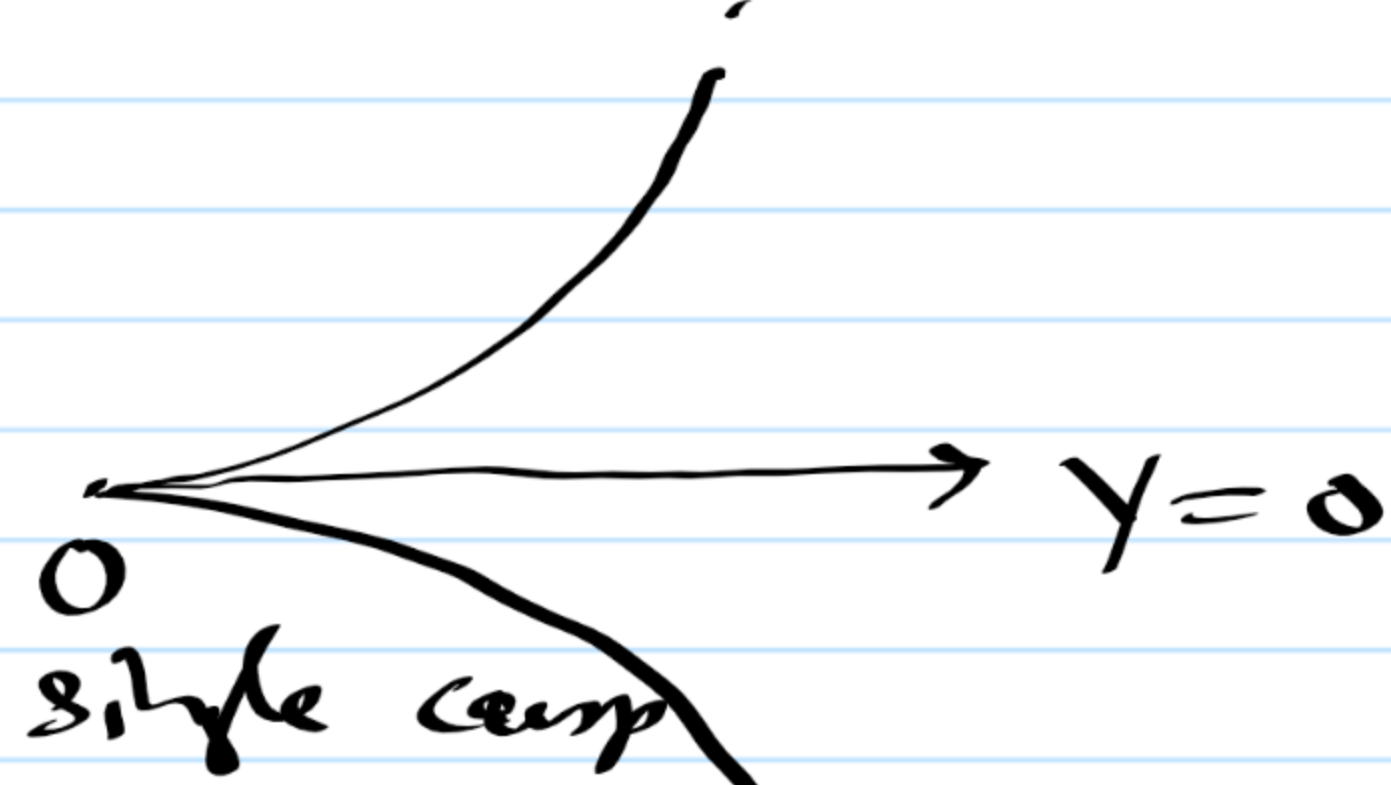
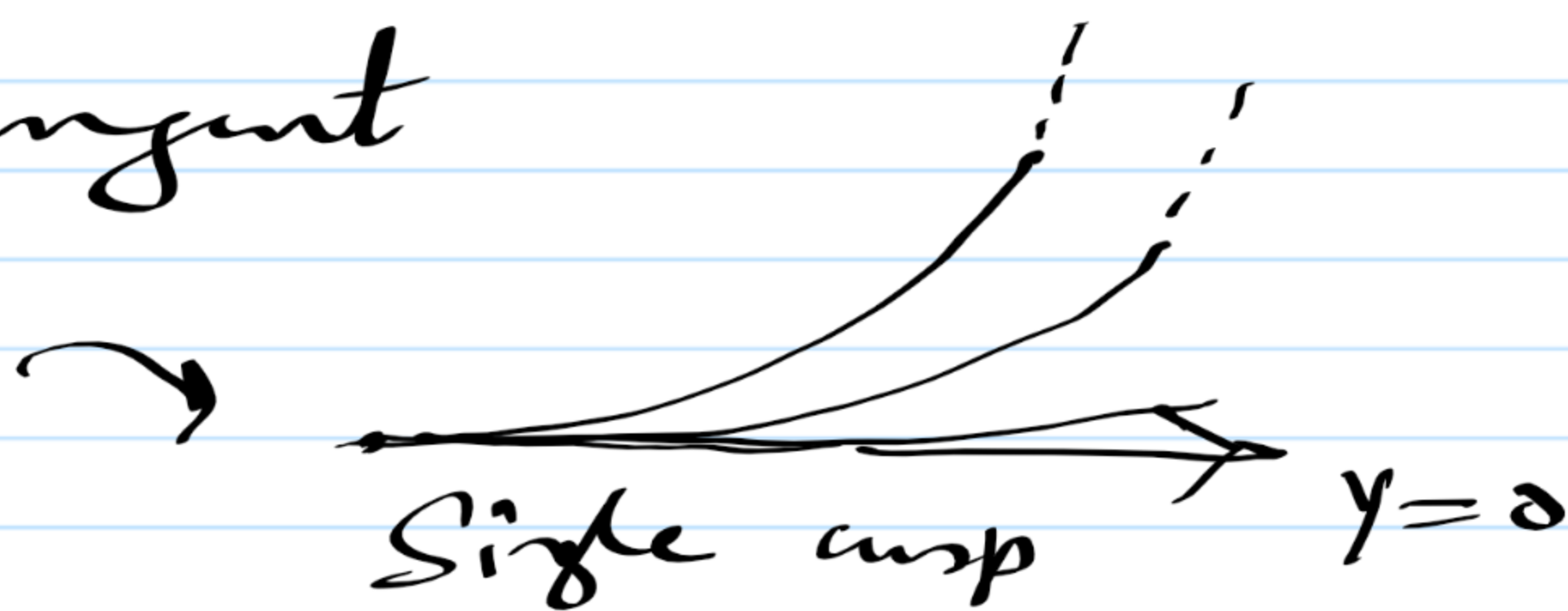
Types of Singular pts : • Node : A singular pt where there are two branches with two distinct tangents.

$$x^3 + y^3 + 3xy = 0$$



Cusp : A double pt, i.e. a singular pt with two branches through it with the same tangent

1st species



If both branches are on the same side of the tangent at the cusp, then it is of 1st species.

If the two branches are on different sides of the tangent at the cusp, then it is of 2nd species.

Point of Oscul-inflexion: A double cusp, is called
if: at least one ~~the one~~ branch is on both sides of the tangent at the cusp to the curve.

Ex. 1 $f = x^3 y^3 + x^2$

$$f_y = f_x = 0 \quad 3x^2 + 2x = 0 \quad \& \quad 3y^2 = 0 \Rightarrow y = 0$$

$$x = 0 \text{ or } -\frac{3}{2}$$

$$\left(-\frac{3}{2}\right)^3 + \left(-\frac{3}{2}\right)^2 \neq 0 \quad \text{so} \quad (0,0) \text{ is a singular pt.}$$

$x=0 \therefore x=0$, the axis is the tangent to it.

Conditions for a double (singular) pt. for $f(x,y)=0$

✓ $f_x(x_0, y_0) = 0 = f_y(x_0, y_0) \quad \& \quad f(x_0, y_0) = 0$

at $p(x_0, y_0)$ $\left(\frac{\partial^2 f}{\partial x \partial y}\right)^2(p) - \left(\frac{\partial^2 f}{\partial x^2}\right)_p \left(\frac{\partial^2 f}{\partial y^2}\right)_p > 0$

Note

$\left(\frac{\partial^2 f}{\partial x \partial y}\right)^2(p) - \left(\frac{\partial^2 f}{\partial x^2}\right)_p \left(\frac{\partial^2 f}{\partial y^2}\right)_p = 0$ cusp.

$\left(\frac{\partial^2 f}{\partial x \partial y}\right)^2(p) - \left(\frac{\partial^2 f}{\partial x^2}\right)_p \left(\frac{\partial^2 f}{\partial y^2}\right)_p < 0$ conjugate pt (isolated pt)

from at least one of $\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial x \partial y}, \frac{\partial^2 f}{\partial y^2}$ here $\frac{\partial^2 f}{\partial x \partial y} \neq 0$ non-zero

• If $\frac{\partial^2 f}{\partial x^2}(p) = \frac{\partial^2 f}{\partial y^2}(p) = \frac{\partial^2 f}{\partial x \partial y}(p) = 0$, a singularity
 that triple or higher

Ex: $f(x,y) = \underline{x^2(x-y)} + y^2 = x^3 - x^2y + y^2 = 0$

$f_x = f_y = 0 \quad \underline{3x^2 - 2xy} = 0 \quad \& \quad \underline{-x^2 + 2y} = 0 \quad \begin{matrix} (x=2y) \\ y = x/2 \end{matrix}$

$3x^2 - 2x \times \frac{x}{2} \Rightarrow x^2 = 0, \quad 3 - x = 0 \Rightarrow x = 0, 3$
 $\& \quad y = 0, \frac{3}{2}$

$\therefore (0,0) \& (3, \frac{3}{2})$ are possible singularities

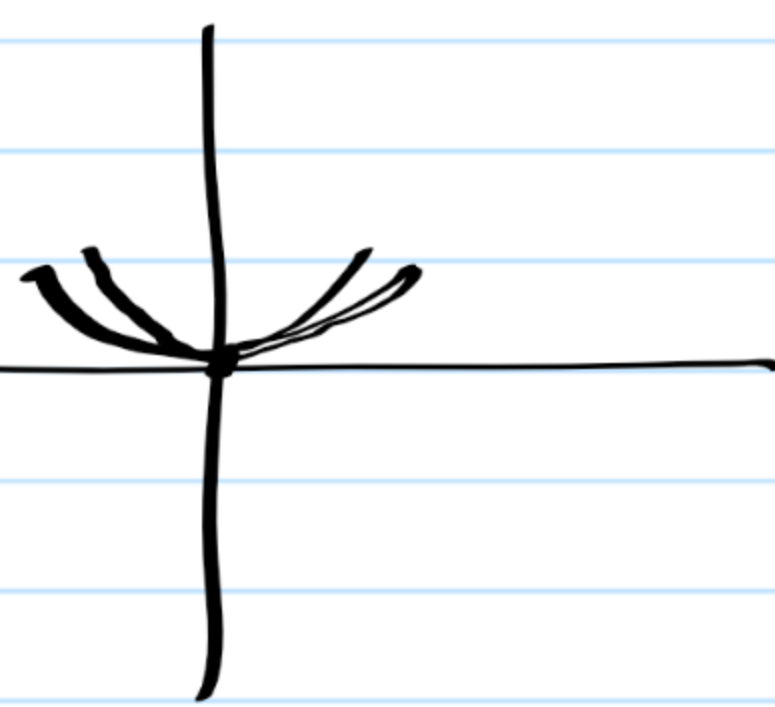
Verify $f(3, \frac{3}{2}) \neq 0$. $(0,0)$ is a singularity.

$y^2 = 0$ are tangents
 $\therefore y=0, y=0$

$\frac{\partial^2 f}{\partial x \partial y}(0) + \frac{\partial^2 f}{\partial x^2}(0) \frac{\partial^2 f}{\partial y^2} = (-2x)|_{x=0} - (6x - 2y)|_{x=0, y=0} \times 2$
 $= 0$

Observe $f(0,y) = y^2$

rather $f(\varepsilon, y) = \underbrace{\varepsilon^2(\varepsilon - y)}_{\text{Very small}} + y^2$



Ex 2

$A, B \in \mathbb{R}$ fixed.

Theorem: let $E = E_{\mathbb{R}} = \{(x,y) \in \mathbb{R}^2 \mid y^2 = x^3 + Ax + B\} \cup \{\infty\}$

Then the following statements are equivalent:

(1) E is non singular.

(2) $g(x) = x^3 + Ax + B$ has no repeated root

(3) $4A^3 + 27B^2 \neq 0$

Set $f = y^2 - x^3 - Ax - B$

Proof 1 $f_x = 0 = f_y \Rightarrow -3x^2 - A = 0 \text{ \& } y = 0$

$$x^2 = -\frac{A}{3} \quad y = 0$$

$$x_0 = \pm \sqrt{-A/3}, \quad y_0 = 0$$

Observe that $g'(x_0) = 0 \checkmark$

If (x_0, y_0) is on E , then $g(x_0) = 0 \checkmark$ as $y_0 = 0$.

i.e. x_0 is a repeated root of $g(x)$.

So $(2) \Rightarrow (1)$

For $(1) \Rightarrow (2)$ if x is a repeated root of $g(x) = 0$
then $(x_0, 0)$ is a singular pt.

$(2) \Rightarrow (2)$ Suppose $g(x)$ has a repeated root

Case (1) $g(x) = (x - \alpha)^3$ (multiplicity = 3)

$$= x^3 - 3\alpha x^2 + 3\alpha^2 x - \alpha^3$$

Compare with $x^3 + Ax + B$.

$$-3\alpha = 0, \quad A = 3\alpha^2, \quad B = -\alpha^3$$

$$\underline{\alpha = 0} \quad \underline{A = B = 0}$$

Case (2) $g(x) = (x - \alpha)^2 (x - \beta)$ $\beta \neq \alpha$

$$= x^3 - (\beta + 2\alpha)x^2 + (\alpha^2 + 2\alpha\beta)x - \alpha^2\beta$$

Compare with $x^3 + Ax + B$

$$\boxed{\beta + 2\alpha = 0} \rightarrow \begin{aligned} A &= \alpha^2 + 2\alpha\beta \neq \beta = -\alpha^2\beta \\ &= \alpha^2 - 4\alpha^2 \\ &= -3\alpha^2 \end{aligned} \quad \underline{B = -\alpha^2\beta} = 2\alpha^3$$

$$\begin{aligned} 4A^3 + 27B^2 &= 4 \times (-3\alpha^2)^3 + 27(2\alpha^3)^2 \\ &= 27 \times (-\alpha^6 + \alpha^6) \\ &= 0 \end{aligned}$$

Q for the converse i.e. (2) $\Rightarrow$ (5)

Assume $4A^3 + 27B^2 = 0$

& reverse the argument to
set α, β .